Questions

Q1.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which of these forces keeps the Moon in orbit around the Earth?

(1)

☐ **A** gravitational
☐ **B** friction
☐ **C** electrostatic
☐ **D** contact

(Total for question = 1 mark)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 19 shows four forces, P, Q, R and S, acting on a rod.

The rod can rotate around an axle.

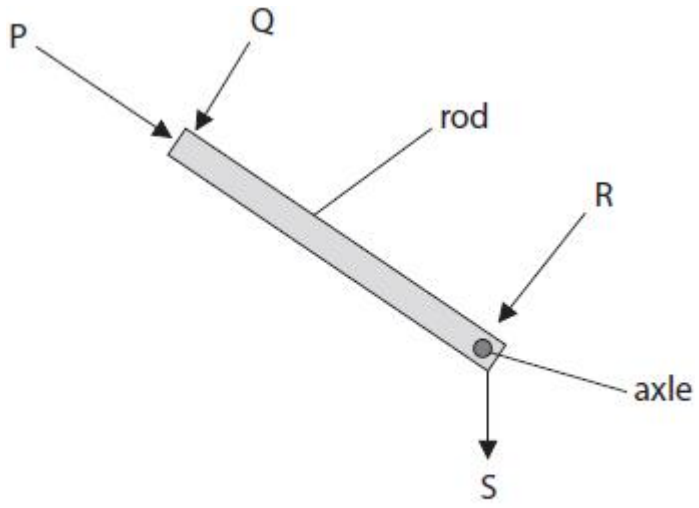

Figure 19

Which force will make the rod rotate about the axle?

(1)

☐ **A** P
☐ **B** Q
☐ **C** R
☐ **D** S

(Total for question = 1 mark)

Q3.

Figure 6 shows two gear wheels, P and Q.

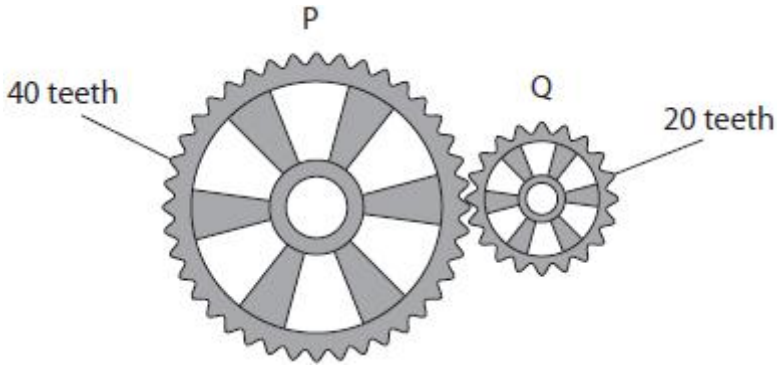

Figure 6

When P turns, it causes Q to turn.

P makes one complete clockwise turn.

Which of these shows the effect on Q?

(1)

- ☐ **A** Q makes half a complete clockwise turn
- ☐ **B** Q makes 2 complete clockwise turns
- ☐ **C** Q makes half a complete anticlockwise turn
- ☐ **D** Q makes 2 complete anticlockwise turns

(Total for question = 1 mark)

Q4.

(i) Which of these forces keeps the Moon moving around the Earth?

(1)

- ☐ **A** contact
- ☐ **B** electrostatic
- ☐ **C** gravitational
- ☐ **D** magnetic

(ii) Which of these is a scalar quantity?

(1)

- ☐ **A** velocity

☐ **B** momentum
☐ **C** energy
☐ **D** acceleration

(Total for question = 2 marks)

Q5.

A student places a trolley at the top of a runway, as shown in Figure 17.

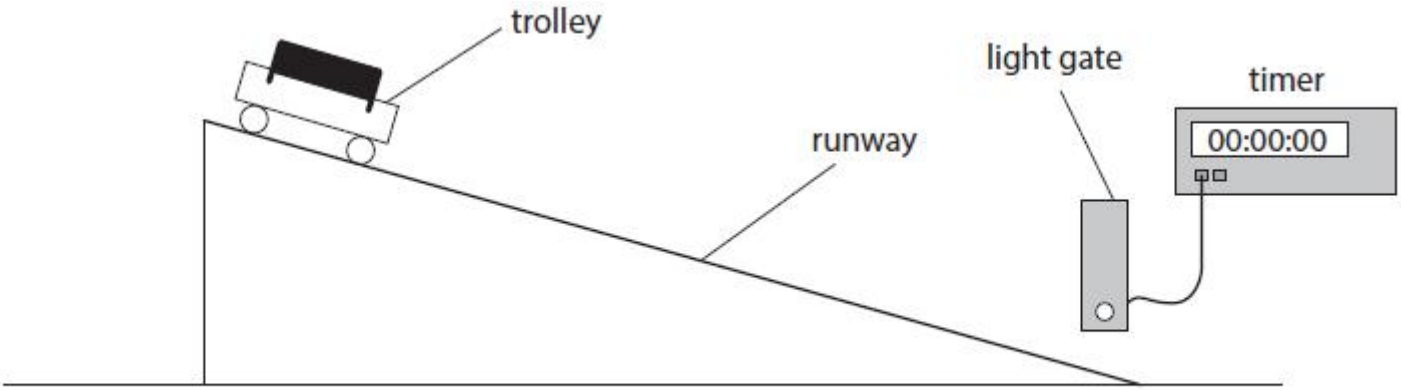

Figure 17

The student had to lift the trolley through a height of 26 cm to put it at the top of the runway.

The gravitational potential energy, ΔGPE, of the trolley increases by 4.1 J when lifted to the top of the runway.

The gravitational field strength, g, = 10 N/kg.

(i) Show that the mass of the trolley is about 1.6 kg.

Use the equation

$$m = \frac{\Delta GPE}{g \times \Delta h}$$

(2)

(ii) The student releases the trolley and the trolley moves down the runway.

The student assumes that all the extra gravitational potential energy, ΔGPE, stored at the top of the runway is transferred to kinetic energy at the end of the runway.
Calculate the expected value of the velocity of the trolley at the end of the runway.
Use the equation

$$v^2 = \frac{2 \times \Delta GPE}{m}$$

(3)

expected velocity = ... m/s

(iii) The light gate, shown in Figure 17, is used to determine the actual velocity of the trolley at the end of the runway.

The actual velocity at the end of the runway is less than the expected velocity as calculated in part (ii). Suggest **one** way to reduce the difference between the actual velocity and the expected velocity.

(1)

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(Total for question = 6 marks)

Q6.

Figure 20 shows a person trying to lift a large rock using a metal bar.

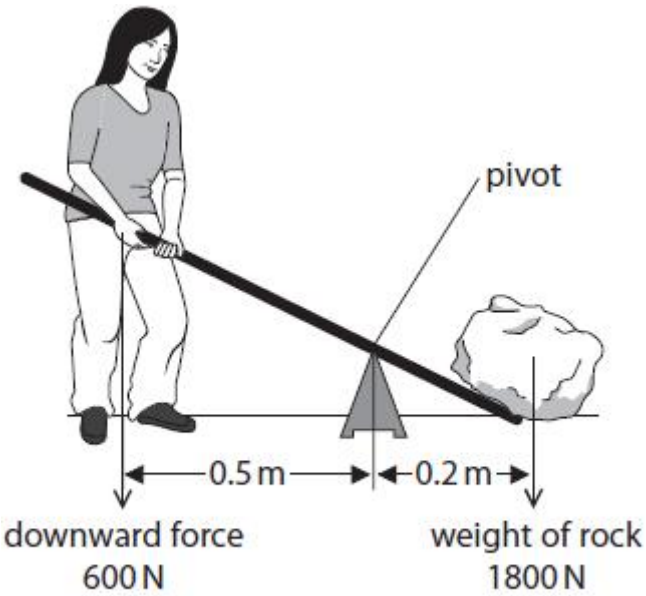

Figure 20

The rock weighs 1800 N.

The person can only produce a downwards force of 600 N.

The person cannot lift the rock.

(i) Explain, using calculations, why the person cannot lift the rock.

(3)

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(ii) Explain **one** change to the arrangement that will make it possible for this person to lift the rock.

(2)

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(Total for question = 5 marks)

Q7.

* Figure 19 shows a crowbar being used to lift a heavy weight.

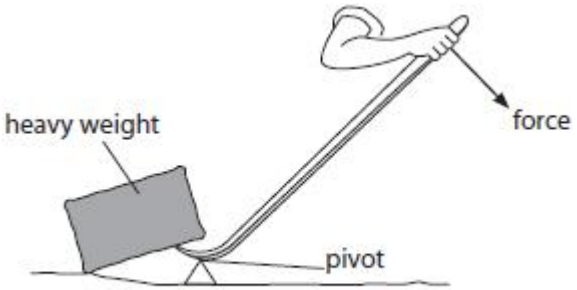

Figure 19

Explain how the crowbar enables a person to lift a heavy weight.

You should include the idea of moments in your answer.

It may help to label some distances in Figure 19 and use those distances in your explanation.

(6)

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(Total for question = 6 marks)

Q8.

Figure 23 shows a drone.

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Figure 23

The drone has four spinning blades.

The upward force produced enables the drone to rise in the air.

The speed at which the blades spin is measured in turns per minute.

Figure 24 shows how the upward force produced by the four blades depends on the speed at which the

blades spin.

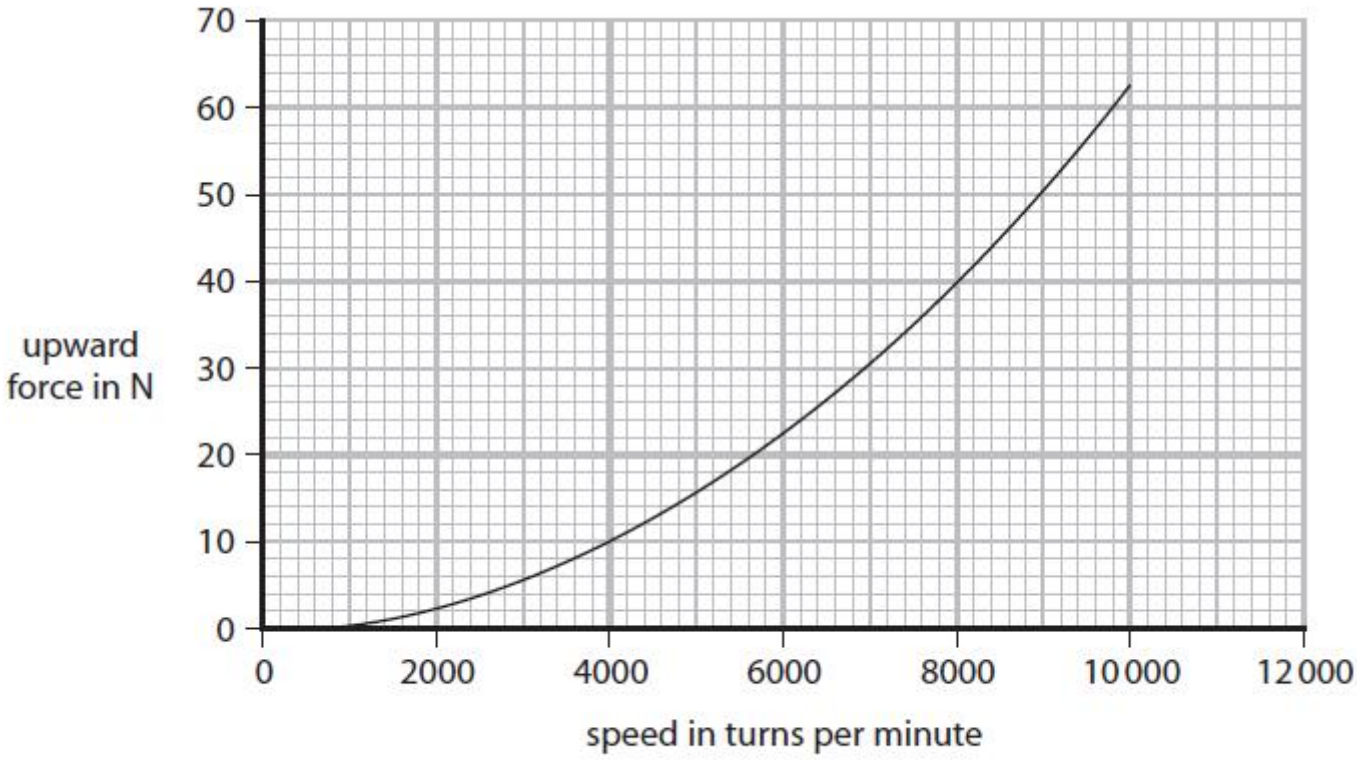

Figure 24

Describe the relationship between upward force and speed shown by this graph.

(2)

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(Total for question = 2 marks)

Q9.

Some forces act at a distance.

One example is the gravitational attraction between the Moon and the Earth.

Describe an example of another type of force acting at a distance, where the force is **not** gravitational.

(2)

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(Total for question = 2 marks)

Q10.

Complete the following sentences using words from the box below.

electrostatic	friction	gravitational	magnetic

(i) The force that keeps the Earth in orbit around the sun is
..... attraction.

(1)

(ii) The force that opposes motion between two surfaces is
..... .

(1)

(iii) The force that can move small pieces of paper towards a plastic comb
is ..... attraction.

(1)

(Total for question = 3 marks)

Mark Scheme

Q1.

Question Number	Answer	Mark
	A gravitational **B,C** and **D** are incorrect because the only force that keeps the Moon in orbit around the Earth is gravitational	**(1)** **AO1.1**

Q2.

Question Number	Answer	Mark
	The only correct answer is B: force Q **A is incorrect** because the moment of force P about the axle is zero. **C is incorrect** because moment of force R about the axle is zero. **D is incorrect** because moment of force S about the axle is zero.	**(1)**

Q3.

Question number	Answer	Mark
	D makes 2 complete anticlockwise turns	**(1)**

Q4.

Question Number:	Answer	Mark
(i)	C gravitational **The only correct answer is C** ***A*** *is not correct as the moon does not touch the Earth* ***B*** *is not correct as the Earth does not carry a charge* ***D*** *is not correct as the Earth has a magnetic field but it does not extend far enough to have any effect on the moon*	**(1)** AO 2 1

Question Number:	Answer	Mark
(ii)	C energy **The only correct answer is C** ***A*** *is not correct as velocity is a vector quantity* ***B*** *is not correct as momentum is a vector quantity* ***D*** *is not correct as acceleration is a vector quantity*	**(1)** AO 1 1

Q5.

Question Number	Answer	Additional guidance	Mark
(i)	substitution with conversion (2) (m =) $\frac{4.1}{10 \times 0.26}$ (=1.58)	accept 9.8(1) instead of 10 for g throughout alternative method substitution and rearrangement with conversion (2) e.g. Δgpe = 10 x 0.26 x 1.6 (=4.1) Allow for 1 mark: (m =) $\frac{4.1}{10 \times 26}$ allow 26 to any power of 10 for this mark OR Δgpe = 10 x 26 x 1.6 allow 26 to any power of 10 for this mark OR height converted to m 0.26 (m) or 26 x 10^{-2} (m)	**(2)** **AO 2.1**

Question Number	Answer	Additional guidance	Mark
(ii)		ecf from part a(i) throughout	(3) AO2.1
	substitution (1) $(v^2=)\ \frac{2 \times 4.1}{1.6}$	accept use of 1.58	
	evaluation of v^2 (1) 5.1(25)	accept 5.2 (use of 1.58)	
	evaluation of v (1) 2.3 (m/s)	accept values that round to 2.3 e.g. 2.264 award full marks for the correct answer without working	

Question Number	Answer	Additional guidance	Mark
(iii)	addition of any lubricant / oil / grease	make runway steep(er) use an air track instead streamline trolley	(1) AO 3.3b

Q6.

Question Number	Answer	Additional guidance	Mark
(i)	recall of moment = force x distance (1) (moment of force from person =) 600 x 0.5 **and** (moment of weight of rock =) 1800 x 0.2 (1) **moment** of force from person is less than **moment** of weight of rock. (1)	may be implied in a calculation 300 (Nm) 360 (Nm) independent mark accept reverse argument	**(3)**

Question Number	Answer	Additional guidance	Mark
(ii)	An explanation that links increase distance between person and pivot/ reduce distance between rock and pivot / increase force from person (1) increase the moment of the force from the person / decrease the moment of the weight of the rock (1)	 use longer lever / hold lever nearer the end / move pivot nearer to rock / get someone to help to push value of new distance and calculation of new moment	**(2)**

Q7.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. • benefit → easier with a lever (AO2) • crowbar easier to lift/move (AO2) • (applied force) is less (AO1) • distance to pivot from (applied) force is (considerably) bigger than distance of load/weight to pivot (AO2) • labelled distances in figure xx (AO2) • force (applied) x x = load x y i.e. principle of moments used (AO1) • relevant mention of clockwise and anticlockwise moments (AO1) • specific application to crowbar (AO2)	**(6)** **AO1, AO2**

Level	Mark	Descriptor
	0	No awardable content
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance elements of physics understanding with some linking to scientific ideas/practical application two isolated statements	Possible candidate responses easier to lift big distance to pivot you can apply your weight onto the crowbar
Level 2	3–4	Additional guidance mostly relevant physics understanding with application of scientific ideas makes some link between force and distance OR some reference to moments	Possible candidate responses less force needed as there is a bigger distance to pivot OR because of moments you need less force
Level 3	5–6	Additional guidance accurate and relevant physics understanding with detailed application of scientific ideas Some reference to crowbar. makes links between force and distance AND refers to moments	Possible candidate responses If moment of weight = moment of crowbar, then the further away you are, you need less force to move the weight the bigger the distance to the pivot, the less force you need to provide the same moment

Q8.

Question number	Answer	Additional guidance	Mark
	(upward) force increases with speed (1) relationship is non-linear (1)	allow reverse argument changing rate / increases exponentially/ initially no upward force (until 1000 turns per minute)	**(2)**

Q9.

Question number	Answer	Additional guidance	Mark
	An answer that combines the following points of understanding to provide a logical description: named force (acting at a distance) (1) situation (1)	 e.g. magnetic force between two (magnetic) poles	**(2)**

Q10.

Question number	Answer	Mark
	(i) gravitational (1) (ii) friction (1) (iii) electrostatic (1)	**(3)** **AO1**